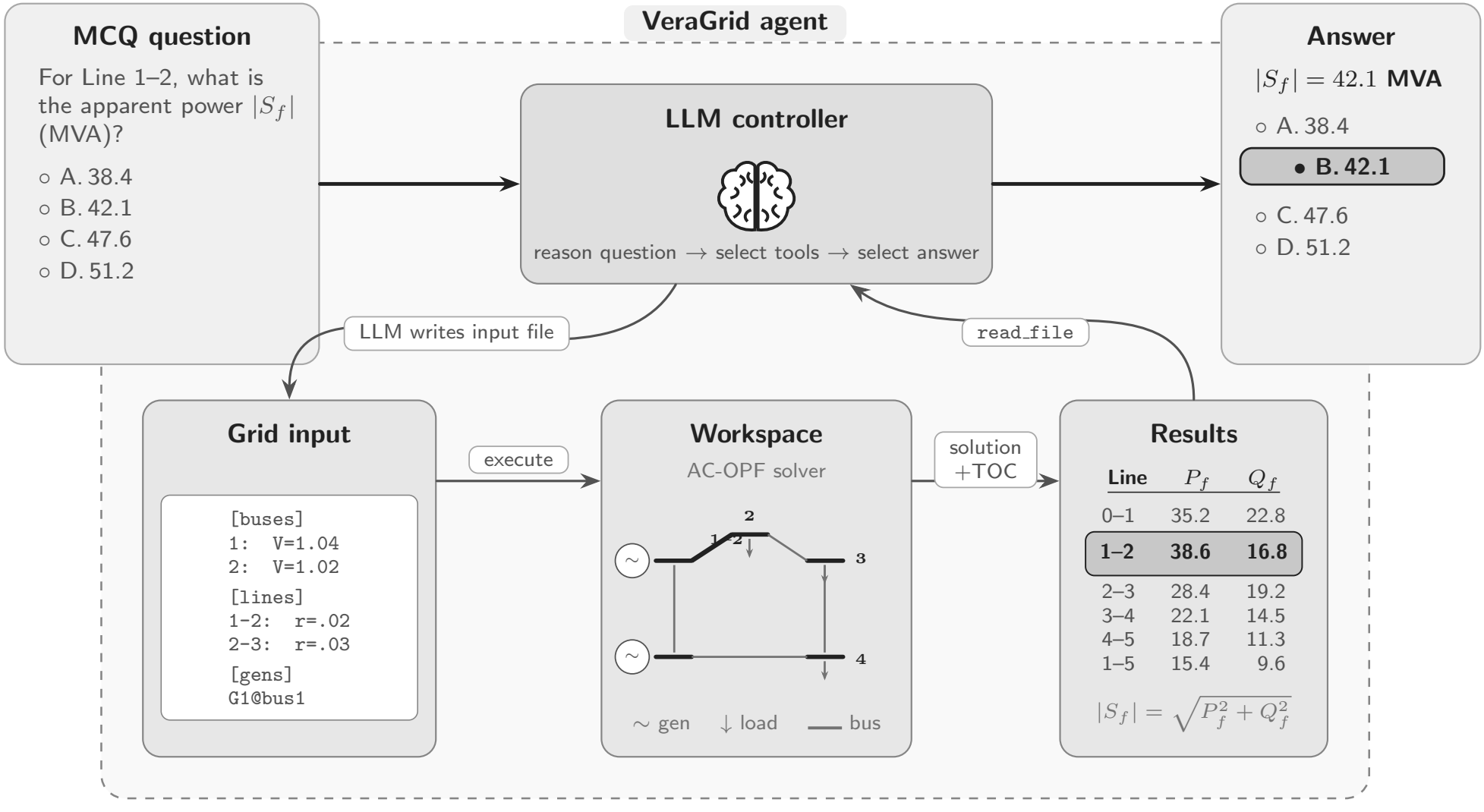

MCQ question
For Line 1–2, what is the apparent power $|S_f|$ (MVA)?
○ A. 38.4
○ B. 42.1
○ C. 47.6
○ D. 51.2
VeraGrid agent
LLM controller
reason question → select tools → select answer
Answer
$|S_f| = 42.1$ MVA
○ A. 38.4
• B. 42.1
○ C. 47.6
○ D. 51.2
LLM writes input file
read_file
Grid input
[buses]
1: V=1.04
2: V=1.02
[lines]
1-2: r=.02
2-3: r=.03
[gens]
G1@bus1
execute
Workspace
AC-OPF solver
1
2
3
4
1–2
~ gen ↓ load — bus
solution +TOC
Results
Line P_f Q_f
0–1 35.2 22.8
1–2 38.6 16.8
2–3 28.4 19.2
3–4 22.1 14.5
4–5 18.7 11.3
1–5 15.4 9.6
$|S_f| = \sqrt{P_f^2 + Q_f^2}$